Guide towards the selection of an appropriate data mining tool

1.0 Introduction

Prediction of the causes of the customer churn in most businesses demands the use of many classifications' algorithms of machine learning for the predictive modeling, as demonstrated in phase two of this project. However, several data mining tools provide a platform for this algorithm (Borges et al., 2013; Kaur & Rana, 2014; SangeethaLakshmi & Jayashree, 2019; Wahbeh et al., 2011; Walia & Arvind, 2020). The examples of the tools listed by them are Waikato Environment for Knowledge Analysis (WEKA), Oracle, Clementine RapidMiner, Rosetta, Tanagra, Intelligent Miner, Orange Canvas, and Konstanz Information Miner (KNIME). In the course of choosing the best tools to address this customer churning with a data mining approach, below is a detailed comparison of the strengths and weaknesses of each tool.

2.0 Comparison of data mining tools

The criteria for selection are cost, functionality such as technique integration, automation in the integration of the mining processes, possible coverage for the Python programming language, and the machine learning libraries. While some tools, such as Oracle and Clementine, are commercialized (Kaur & Rana, 2014; Wahbeh et al., 2011). However, the majority of the tools are open source. In the consideration of the cost implications toward reduction of cost and increment of profit, I will opt to compare open-source tools for the selection of the best to tackle the customer churning facing my financial institution. I am considering four from the pool of open-source data mining tools for the comparison, as deduced from the work of Wahbeh et al. (2011) and Walia & Arvind (2020). These are WEKA, KNIME, Tanagra, and Orange Canvas. Table 1 summarizes the pros and cons of each of the four open sources.

Table 1. Comparison of the strength and weakness of selected open source data mining tools

Criteria	WEKA	Orange	KNIME	Tanagra
Pros	support more input formats like. ARFF, CSV, C4.5, and binary	Support numerous add-ons	Compatibility with all platforms through plugins	Provides architecture to add custom data mining methods.
	Provide interactive GUI	Provide interactive GUI	Suitable for data mining, business analytics, and enterprise reporting	It is less prone to error.
	Support many machine learning algorithms.	Support tools for machine learning.	Offers numerous libraries for classical statistics	It can also serve as a pedagogical tool to learn programming techniques.
	Enable data preparation, feature selection, and data mining.	Enable data visualization of interactive exploration.	It allows integration of other tools, e.g., WEKA for improved mining efficiency.	It blends well with machine learning classical statistics and visualization.
	Easier and portable	Simple and easy to learn	Easy to use	Easy to use
Cons	GUI documentation is limited.	Poor handling of machine learning in diverse libraries	No automatic feature for machine learning hyperparameter optimization	Limited output formats
	Low connectivity for non-Java databases	Low or no compatibility for the Java platform	Inadequate methods for error measurement	
	Traditional statistics are not well incorporated.	Not offering widgets for classical statistics	No wrapper methods for descriptor selection	

3.0 Choice of Konstanz Information Miner over other data mining tools

Having compared the aforementioned tools, I chose KNIME for my company to enhance her activities due to below features in the KNIME.

1. Possible integration with other tools, such as WEKA, and overall business intelligence architecture towards enhancing the data mining efficiency with various machine learning algorithms.

2. Compatibility with all platforms, including many programming languages such as Python used for data mining in phase 2 of this project.

3. Plugins availability to accommodate numerous tasks

4.0 References

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